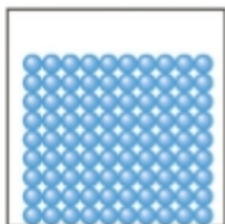


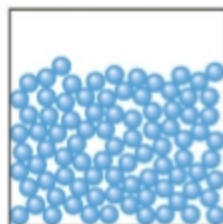
States of matter

Elements can exist in three states: solid, liquid, and gas. An element can change from one state to another. For example, solid gallium can melt into a liquid, while liquid bromine can evaporate into a gas. The changes do not alter the atoms of the element but arrange them in a more, or less, rigid way.



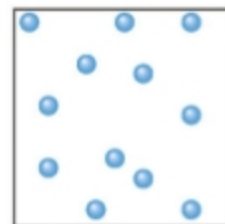
Solid

In solids, the atoms are attracted to one another, are arranged in a regular pattern, and have little energy to move around.



Liquid

As solids become liquids, the attraction between the atoms weakens. They have no fixed arrangement and more energy.



Gas

As liquids become gases, the atoms are very weakly attracted to each other. They spread out as far as possible and have a lot of energy.



Gold in quartz

Pure hydrogen gas in a glass sphere



Raw forms

Elements are commonly found in one of two ways. If they are unreactive and do not easily combine with other elements, they may be found in their pure state, such as gold. Elements that are more reactive always combine with other elements around them. This combination is called a compound.



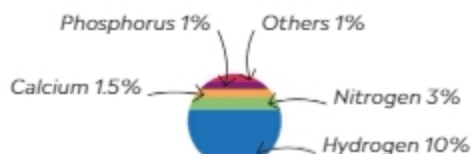
Common salt (sodium chloride)

Compounds

When elements chemically bond with one another in a fixed ratio, they form compounds. For example, in sodium chloride, sodium combines with chlorine in an equal ratio to form a compound. Water is a compound formed when two hydrogen atoms combine with one oxygen atom to make one molecule.

Mixtures

When elements or compounds combine in an unequal ratio without chemically bonding with one another, the resulting combination is called a mixture. Mixtures can be easily separated. For example, the combination of lather and food coloring seen on the right is a mixture that can be separated by filtering.



The human body

About 99 percent of the human body is made from just six elements: oxygen, carbon, hydrogen, nitrogen, calcium, and phosphorus. They combine together to form thousands of different compounds. More than 60 percent of the body is water, but there are about 25 elements that are required to make our bodies work properly. They are called the "essential elements."

